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# Procedural generation: music

Generating music using method X and Y

Master's thesis in Cybernetics and Robotics
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NTNU

# ABSTRACT

Write an abstract/summary of your thesis, and state your main findings here.

A summary should be included in both English and any second language, if this is applicable, regardless if the thesis is written in English or in your preferred language. These should be on separate pages, the English version first.

# PREFACE

Write the preface of your thesis here.

You may include acknowledgements and thanks as part of your preface on this page, or you may add it as a new chapter after the preface.

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# ABBREVIATIONS

List of all abbreviations in alphabetic order:

- **NTNU** Norwegian University of Science and Technology

CHAPTER

# ONE

# INTRODUCTION

For hundreds of years composers have been creating music using external tools like dice to help them generate something new. Creating a good solution for generating music procedurally has yet to be done and this thesis will explore a way to solve this problem.

## 1.1 Motivation

Creating a good solution for generating good music will not just be applicable to generating music, but the structure will be able to be used for a number of other applications. That could be for other artistic purposes or any application where generating something that cannot be completely random.

## 1.2 Project description

Project decs. bla bla

CHAPTER

# TWO

# THEORY

## 2.1 Music theory

Figure 2.1.1 to exemplify music notation as a figure.

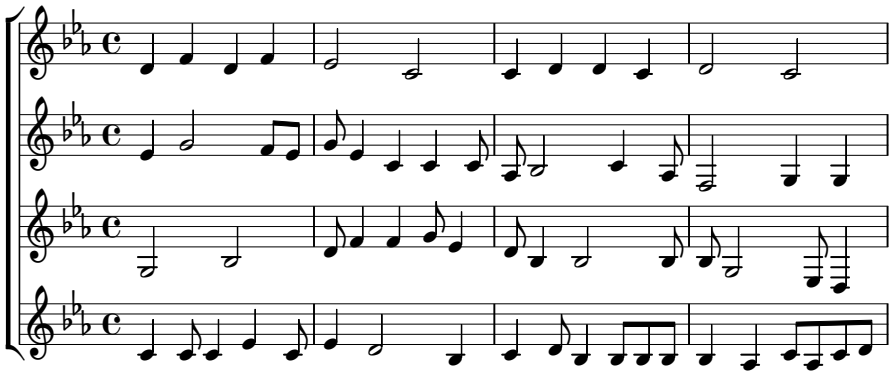

**Figure 2.1.1:** This is a test of displaying notes in a figure. The figure is generated by the code in the file `melody.cropped.png`. The code is located in the folder `output/4-part-random`.

Another figure coming right up.

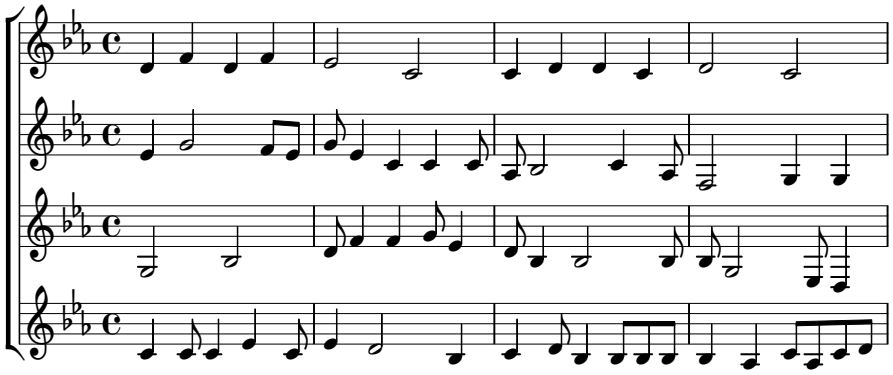

**Figure 2.1.2:** Testing another figure.

CHAPTER

# THREE

# METHODS

Include the complete description of the methods used in your research here.

Below is an example of how subsectioning works. The sections and subsections will be included in the table of contents, while subsubsections will not be in the table of contents but still have their own title in the text.

## 3.1 Section one

### 3.1.1 Subsection one

#### 3.1.1.1 Subsubsection one

#### 3.1.1.2 Subsubsection Two

### 3.1.2 Subsection Two

## 3.2 Section two

## 3.3 Design

CHAPTER

# FOUR

# RESULTS

## 4.1 More figures

CHAPTER

# FIVE

# DISCUSSION

Discuss your results here.

## 5.1 Future work

Include a section about what should or could be done in future research, or explain any recommended next steps based on the results you got. This should be the last section in the discussion.

CHAPTER

# SIX

# CONCLUSIONS

Give a concise summary of your research and finding here, and include a short summary of any future work as well.
Looking at this thing here, [1] [2] [3] it is a citation.

# REFERENCES


[1] Roland van der Linden, Ricardo Lopes, and Rafael Bidarra. "Procedural Generation of Dungeons". In: *IEEE Transactions on Computational Intelligence and AI in Games* 6.1 (Mar. 2014), pp. 78–89. ISSN: 1943-0698. DOI: 10.1109/TCIAIG.2013.2290371. URL: https://ieeexplore.ieee.org/abstract/document/6661386 (visited on 02/26/2026).

[2] E. Galin et al. "Procedural Generation of Roads". In: *Computer Graphics Forum* 29.2 (2010), pp. 429–438. ISSN: 1467-8659. DOI: 10.1111/j.1467-8659.2009.01612.x. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-8659.2009.01612.x (visited on 02/26/2026).

[3] Johan Gangsås Hole. "Automatic Species Counterpoint". In: *Master Thesis at NTNU* (2026).

# APPENDICES

# A - GITHUB REPOSITORY

All code and latex-files used in this document are included in the Github repository linked below. Further explanations are given in the readme-file.

## Github repository link

- `https://github.com/lima98/TTK4900`

# B - AUDIO EXAMPLES

Audio-files for the examples in the thesis can be played from my NTNU-webpage.

## Audio example archive

- `https://www.janoivil.folk.ntnu.no`

# C - CODE DOCUMENTATION

All documentation for the code is included in the Github repository linked in 6 it is also provided as a web-page hosted at my NTNU-webpage.

## Code documentation

- `https://janoivil.folk.ntnu.no/docs/_build/html/index.html`